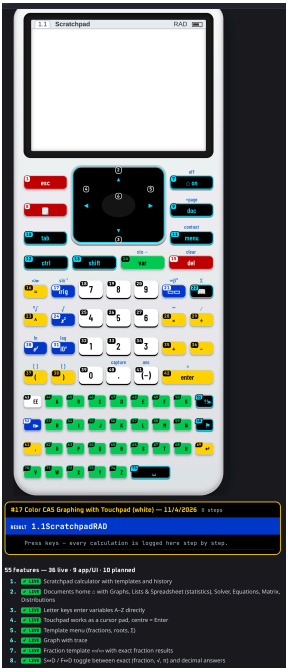
# U-45 CALCULATORS

## User Manual

### Calculator #17 Color CAS Graphing with Touchpad (white)

Graphing – colour




Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

# About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#17 — Color CAS Graphing with Touchpad (white)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

## How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

## Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

## Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)
12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

# 1 · Getting started

Open U-45 CALCULATORS, choose **#17** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

### Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

### High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

### Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

### Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

### Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

### Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

### Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

### Tooltips

Hover over (or long-press) a key to see "#number name".

### Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

### History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

### Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

### Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

### Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

### Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

### Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

## Colour groups on this calculator

| Colour | Group | Keys on this model |
|---|---|---|
| num | Digits & number entry | #18, #19, #20, #25, #26, #27, #32, #33, #34, #39, #40, #41, #43 |
| op | Operators & execute | #6, #16, #23, #28, #29, #35, #36, #37, #38, #42, #61, #69 |
| fn | Functions | #17, #21, #24, #30, #31, #52 |
| mem | Memory, variables & registers | #14, #44, #45, #46, #47, #48, #49, #50, #53, #54, #55, #56, #57, #58, #59, #62, #63, #64, #65, #66, #67, #68, #70, #71, #72, #73, #74 |

| | | |
|---|---|---|
| clr | **Clear, delete & power** | **#1, #8, #15** |
| mod | **Shift, modes, menus & navigation** | **#2, #3, #4, #5, #7, #9, #10, #11, #12, #13, #22, #51, #60, #75** |

# 2 · Features

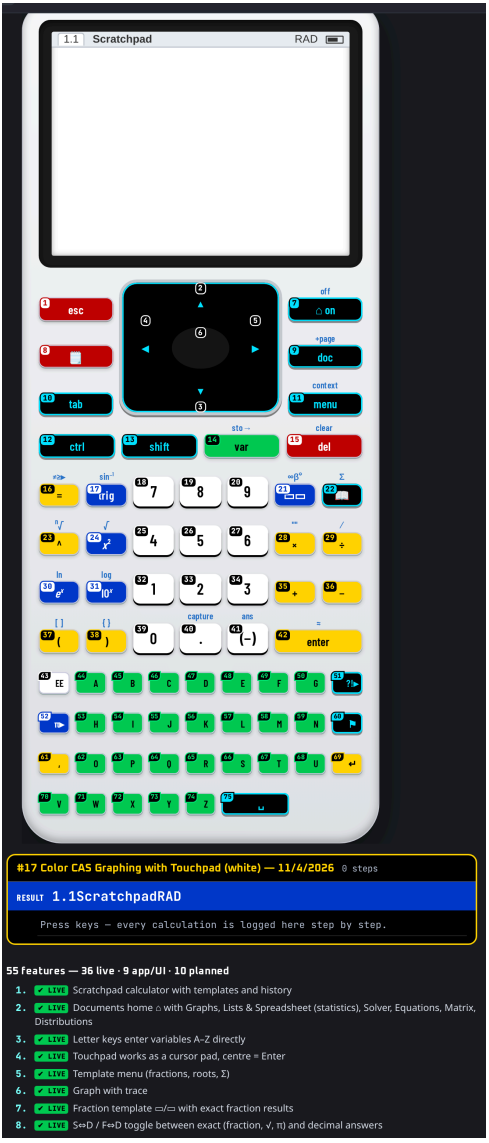
**55 features: 36 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).**

1. LIVE **Scratchpad calculator with templates and history**
2. LIVE **Documents home ⌂ with Graphs, Lists & Spreadsheet (statistics), Solver, Equations, Matrix, Distributions**
3. LIVE **Letter keys enter variables A–Z directly**
4. LIVE **Touchpad works as a cursor pad, centre = Enter**
5. LIVE **Template menu (fractions, roots, Σ)**
6. LIVE **Graph with trace**
7. LIVE **Fraction template ▭/▭ with exact fraction results**
8. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
9. LIVE **Square root √**
10. LIVE **Any-index root $\sqrt[x]{y}$**
11. LIVE **Square $x^2$**
12. LIVE **Power $x^y$ / ^ with right-to-left chaining (2^3^2 = 512)**
13. LIVE **Common logarithm $\log_{10}$**
14. LIVE **Natural logarithm ln**
15. LIVE **Antilog $10^x$**
16. LIVE **Exponential $e^x$**
17. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
18. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**

19. LIVE Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion
20. LIVE π constant to full internal precision
21. LIVE Scientific-notation entry ×10ˣ / EXP / EEX
22. LIVE Sign change (−) / +/− / CHS
23. LIVE Ans recall of the last result; operators auto-insert Ans
24. LIVE Store a value into memory / a variable (STO, sto▸, →, Min)
25. LIVE Named variables (A–F, X, Y, M … or A–Z) usable inside expressions
26. LIVE Summation Σ(f(x), x=a…b)
27. LIVE SOLVE — numeric equation solver (Newton / bisection)
28. LIVE TABLE — tabulate f(x) over start / end / step
29. LIVE Statistics: data entry, n, $\bar{x}$, σ, s, Σx, $\Sigma x^2$
30. LIVE Two-variable statistics and linear regression (a, b, r)
31. LIVE Matrix mode: define, add, multiply, det, transpose, identity
32. LIVE Backspace / DEL editing of the current entry
33. LIVE Replay ▲ ▼ — scroll back to previous calculations and re-edit them
34. LIVE CATALOG — alphabetical list of every function
35. LIVE MATH / Toolbox menu of templates and functions
36. LIVE Probability distributions (normal, binomial, Poisson) and PRB menu
37. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
38. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
39. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)

40. APP/UI Show / hide key numbers with one button (remembered between sessions)
41. APP/UI Copy the current display text to the clipboard
42. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
43. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
44. APP/UI Reference photo viewer and tool-tip name on every key
45. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
46. PLANNED Symbolic CAS algebra
47. PLANNED Documents / pages
48. PLANNED Python
49. PLANNED Touchpad gestures
50. PLANNED Geometry
51. PLANNED Graph image export (PNG)
52. PLANNED Shareable function list link
53. PLANNED Dark screen theme
54. PLANNED Speech read-out of values
55. PLANNED Exam mode

# 3 · Numbered key map




Calculator #17 with key numbers shown (Hi-Contrast colours).

# Key table

**“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.**

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 1 | esc | · | · | · | clr |
| 2 | ▲ | · | · | · | mod |
| 3 | ▼ | · | · | · | mod |
| 4 | ◀ | · | · | · | mod |
| 5 | ▶ | · | · | · | mod |
| 6 | exe | · | · | · | op |
| 7 | ⌂ on | off | · | · | mod |
| 8 | 🗒 | · | · | · | clr |
| 9 | doc | +page | · | · | mod |
| 10 | tab | · | · | · | mod |
| 11 | menu | context | · | · | mod |
| 12 | ctrl | · | · | · | mod |
| 13 | shift | · | · | · | mod |
| 14 | var | sto → | · | · | mem |
| 15 | del | clear | · | · | clr |
| 16 | = | ≠≥▸ | · | · | op |
| 17 | trig | $\sin^{-1}$ | · | · | fn |
| 18 | 7 | · | · | · | num |
| 19 | 8 | · | · | · | num |
| 20 | 9 | · | · | · | num |
| 21 | ▭▭ | ∞β° | · | · | fn |
| 22 | 📖 | Σ | · | · | mod |
| 23 | ^ | n√ | · | · | op |

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 24 | $x^2$ | √ | · | · | fn |
| 25 | 4 | · | · | · | num |
| 26 | 5 | · | · | · | num |
| 27 | 6 | · | · | · | num |
| 28 | × | "" | · | · | op |
| 29 | ÷ | / | · | · | op |
| 30 | ex | ln | · | · | fn |
| 31 | 10x | log | · | · | fn |
| 32 | 1 | · | · | · | num |
| 33 | 2 | · | · | · | num |
| 34 | 3 | · | · | · | num |
| 35 | + | · | · | · | op |
| 36 | − | · | · | · | op |
| 37 | ( | [ ] | · | · | op |
| 38 | ) | { } | · | · | op |
| 39 | 0 | · | · | · | num |
| 40 | . | capture | · | · | num |
| 41 | (−) | ans | · | · | num |
| 42 | enter | ≈ | · | · | op |
| 43 | EE | · | · | · | num |
| 44 | A | · | · | · | mem |
| 45 | B | · | · | · | mem |
| 46 | C | · | · | · | mem |
| 47 | D | · | · | · | mem |
| 48 | E | · | · | · | mem |
| 49 | F | · | · | · | mem |
| 50 | G | · | · | · | mem |
| 51 | ?!▸ | · | · | · | mod |

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 52 | π▶ | · | · | · | fn |
| 53 | H | · | · | · | mem |
| 54 | I | · | · | · | mem |
| 55 | J | · | · | · | mem |
| 56 | K | · | · | · | mem |
| 57 | L | · | · | · | mem |
| 58 | M | · | · | · | mem |
| 59 | N | · | · | · | mem |
| 60 | ⚑ | · | · | · | mod |
| 61 | , | · | · | · | op |
| 62 | O | · | · | · | mem |
| 63 | P | · | · | · | mem |
| 64 | Q | · | · | · | mem |
| 65 | R | · | · | · | mem |
| 66 | S | · | · | · | mem |
| 67 | T | · | · | · | mem |
| 68 | U | · | · | · | mem |
| 69 | ↵ | · | · | · | op |
| 70 | V | · | · | · | mem |
| 71 | W | · | · | · | mem |
| 72 | X | · | · | · | mem |
| 73 | Y | · | · | · | mem |
| 74 | Z | · | · | · | mem |
| 75 | ␣ | · | · | · | mod |

# 4 · Key-by-key reference

**Every key on calculator #17, in number order, with every function it carries in this app.**

### #1 esc `Clear, delete & power`

**Primary:** QUIT — leave the current screen / app

### #2 ▲ `Shift, modes, menus & navigation`

**Primary:** Cursor ▲ — up / previous history line / previous item

### #3 ▼ `Shift, modes, menus & navigation`

**Primary:** Cursor ▼ — down / next history line / next item

### #4 ◀ `Shift, modes, menus & navigation`

**Primary:** Cursor ◀ — move left in the line

### #5 ▶ `Shift, modes, menus & navigation`

**Primary:** Cursor ▶ — move right in the line

### #6 exe `Operators & execute`

**Primary:** Execute / = — evaluate the line and show the result (EXE / ENTER / =)

### #7 ⌂ on `Shift, modes, menus & navigation`

**Primary:** APPS — applications

**shifted (2nd / f / SHIFT layer) (press #12 first):** off — OFF — power off (state is lost; press ON to restart)

### #8 🗒️ `Clear, delete & power`

**Primary:** QUIT — leave the current screen / app

### #9 doc `Shift, modes, menus & navigation`

**Primary:** APPS — applications

### #10 tab `Shift, modes, menus & navigation`

**Primary:** Cursor ▼ — down / next history line / next item

**Keyboard:** `ArrowDown`

### #11 menu `Shift, modes, menus & navigation`

**Primary:** MATH menu — extra math functions

**shifted (2nd / f / SHIFT layer) (press #12 first):** context — OPTN — options for the current mode

### #12 ctrl `Shift, modes, menus & navigation`

**Primary:** SHIFT / 2nd — the next key uses its upper (shift-coloured) function

## #13 shift `Shift, modes, menus & navigation`

**Primary:** Printed label only (no action in this app)

## #14 var `Memory, variables & registers`

**Primary:** VARS — variables menu

**shifted (2nd / f / SHIFT layer) (press #12 first):** sto → — STO▶ ( → ) — store into a variable

## #15 del `Clear, delete & power`

**Primary:** DEL — delete the character at the cursor

**shifted (2nd / f / SHIFT layer) (press #12 first):** clear — CLEAR — clear the entry line / screen

**Keyboard:** `Backspace` `Delete`

## #16 = `Operators & execute`

**Primary:** Equals — complete the calculation

**shifted (2nd / f / SHIFT layer) (press #12 first):** ≠≥▶ — Types "≠"

## #17 trig `Functions`

**Primary:** Sine (uses the current angle unit)

**shifted (2nd / f / SHIFT layer) (press #12 first):** $\sin^{-1}$ — Inverse sine $\sin^{-1}$

**Keyboard:** `s`

**#18** **7** `Digits & number entry`

**Primary:** Digit 7

**Keyboard:** 7

**#19** **8** `Digits & number entry`

**Primary:** Digit 8

**Keyboard:** 8

**#20** **9** `Digits & number entry`

**Primary:** Digit 9

**Keyboard:** 9

**#21** ▭▭ `Functions`

**Primary:** Fraction template ▭/▭ (numerator over denominator)

**shifted (2nd / f / SHIFT layer) (press #12 first):** ∞β° — Degrees-minutes-seconds entry/display °′ ″

**#22** 📖 `Shift, modes, menus & navigation`

**Primary:** CATALOG — list of all commands

**shifted (2nd / f / SHIFT layer) (press #12 first):** Σ — Summation Σ over an integer range

## #23 ^ Operators & execute

**Primary:** Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

**shifted (2nd / f / SHIFT layer) (press #12 first):** $\sqrt[n]{}$ — n-th root $\sqrt[x]{y}$

**Keyboard:** [^]

## #24 $x^2$ Functions

**Primary:** Square $x^2$

**shifted (2nd / f / SHIFT layer) (press #12 first):** $\sqrt{}$ — Square root $\sqrt{}$

**Keyboard:** [q]

## #25 4 Digits & number entry

**Primary:** Digit 4

**Keyboard:** [4]

## #26 5 Digits & number entry

**Primary:** Digit 5

**Keyboard:** [5]

## #27 6 Digits & number entry

**Primary:** Digit 6

**Keyboard:** [6]

**#28** × Operators & execute

**Primary:** Multiplication ×

**Keyboard:** * x

**#29** ÷ Operators & execute

**Primary:** Division ÷

**shifted (2nd / f / SHIFT layer) (press #12 first):** / — Fraction template ▭/▭ (numerator over denominator)

**Keyboard:** /

**#30** ex Functions

**Primary:** $e^x$ — exponential

**shifted (2nd / f / SHIFT layer) (press #12 first):** ln — Natural logarithm ln

**#31** 10x Functions

**Primary:** $10^x$ — antilog

**shifted (2nd / f / SHIFT layer) (press #12 first):** log — Common logarithm $\log_{10}$

**#32** 1 Digits & number entry

**Primary:** Digit 1

**Keyboard:** 1

**#33** **2** Digits & number entry

**Primary:** Digit 2

**Keyboard:** 2

**#34** **3** Digits & number entry

**Primary:** Digit 3

**Keyboard:** 3

**#35** **+** Operators & execute

**Primary:** Addition +

**Keyboard:** +

**#36** **–** Operators & execute

**Primary:** Subtraction –

**Keyboard:** -

**#37** **(** Operators & execute

**Primary:** Opening bracket (

**shifted (2nd / f / SHIFT layer) (press #12 first):** [ ] — Types "["

**Keyboard:** (

### #38 ) Operators & execute

**Primary:** Closing bracket ). Closing brackets at the end of a line may be omitted

**shifted (2nd / f / SHIFT layer) (press #12 first):** { } — Types "{"

**Keyboard:** )

### #39 0 Digits & number entry

**Primary:** Digit 0

**Keyboard:** 0

### #40 . Digits & number entry

**Primary:** Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

**Keyboard:** .

### #41 (–) Digits & number entry

**Primary:** Negative sign (–) — makes the next number negative. Different from the subtraction key

**shifted (2nd / f / SHIFT layer) (press #12 first):** ans — Ans — recall the last answer into the current expression

## #42 enter Operators & execute

**Primary:** Execute / = — evaluate the line and show the result (EXE / ENTER / =)

**shifted (2nd / f / SHIFT layer) (press #12 first):** ≈ — ▶Decimal — show the result as a decimal

**Keyboard:** Enter =

## #43 EE Digits & number entry

**Primary:** ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

**Keyboard:** e

## #44 A Memory, variables & registers

**Primary:** Variable A (store with STO/ →, recall by name)

## #45 B Memory, variables & registers

**Primary:** Variable B (store with STO/ →, recall by name)

## #46 C Memory, variables & registers

**Primary:** Variable C (store with STO/ →, recall by name)

## #47 D Memory, variables & registers

**Primary:** Variable D (store with STO/ →, recall by name)

### #48 E `Memory, variables & registers`

**Primary:** Variable E (store with STO/ →, recall by name)

### #49 F `Memory, variables & registers`

**Primary:** Variable F (store with STO/ →, recall by name)

### #50 G `Memory, variables & registers`

**Primary:** Variable G (store with STO/ →, recall by name)

### #51 ?!▶ `Shift, modes, menus & navigation`

**Primary:** Printed label only (no action in this app)

### #52 π▶ `Functions`

**Primary:** π (3.14159265…) as an exact constant

**Keyboard:** p

### #53 H `Memory, variables & registers`

**Primary:** Variable H (store with STO/ →, recall by name)

### #54 I `Memory, variables & registers`

**Primary:** Variable I (store with STO/ →, recall by name)

**#55** **J** `Memory, variables & registers`

**Primary:** Variable J (store with STO/ →, recall by name)

**#56** **K** `Memory, variables & registers`

**Primary:** Variable K (store with STO/ →, recall by name)

**#57** **L** `Memory, variables & registers`

**Primary:** Variable L (store with STO/ →, recall by name)

**#58** **M** `Memory, variables & registers`

**Primary:** Variable M (store with STO/ →, recall by name)

**#59** **N** `Memory, variables & registers`

**Primary:** Variable N (store with STO/ →, recall by name)

**#60** ⚑ `Shift, modes, menus & navigation`

**Primary:** Printed label only (no action in this app)

**#61** **,** `Operators & execute`

**Primary:** Comma / argument separator (between function arguments, list items or coordinates)

**Keyboard:** [,]

**#62 O** `Memory, variables & registers`

**Primary: Variable O (store with STO/ →, recall by name)**

**#63 P** `Memory, variables & registers`

**Primary: Variable P (store with STO/ →, recall by name)**

**#64 Q** `Memory, variables & registers`

**Primary: Variable Q (store with STO/ →, recall by name)**

**#65 R** `Memory, variables & registers`

**Primary: Variable R (store with STO/ →, recall by name)**

**#66 S** `Memory, variables & registers`

**Primary: Variable S (store with STO/ →, recall by name)**

**#67 T** `Memory, variables & registers`

**Primary: Variable T (store with STO/ →, recall by name)**

**#68 U** `Memory, variables & registers`

**Primary: Variable U (store with STO/ →, recall by name)**

**#69** ↵ `Operators & execute`

**Primary:** Execute / = — evaluate the line and show the result (EXE / ENTER / =)

**#70** V `Memory, variables & registers`

**Primary:** Variable V (store with STO/ →, recall by name)

**#71** W `Memory, variables & registers`

**Primary:** Variable W (store with STO/ →, recall by name)

**#72** X `Memory, variables & registers`

**Primary:** Variable X (store with STO/ →, recall by name)

**#73** Y `Memory, variables & registers`

**Primary:** Variable Y (store with STO/ →, recall by name)

**#74** Z `Memory, variables & registers`

**Primary:** Variable Z (store with STO/ →, recall by name)

**#75** ␣ `Shift, modes, menus & navigation`

**Primary:** Printed label only (no action in this app)

# 5 · Worked examples

**16 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.**

### Example 5.1 — Order of operations

**Calculate 12 + 3 × 4. Press Reset first.**

#32 1 → #33 2 → #35 + → #34 3 → #28 × → #25 4 → #6

**Display shows 24 — expected 24. ✔ verified on the app.**

*Multiplication is done before addition, so the calculator holds the + until the × part is finished.*

### Example 5.2 — Brackets first

**Calculate (12 + 8) ÷ 4. Press Reset first.**

#37 ( → #32 1 → #33 2 → #35 + → #19 8 → #38 ) → #29 ÷ → #25 4 → #6

**Display shows 5 — expected 5. ✔ verified on the app.**

*Brackets force the sum to be worked out before the division.*

### Example 5.3 — Two brackets multiplied

Calculate **(3 + 4) × (9 − 2)**. Press **Reset** first.

#37 ( → #34 3 → #35 + → #25 4 → #38 ) → #28 × → #37 ( → #20 9 → #36 − → #33 2 → #38 ) → #6

Display shows 49 — expected 49. ✔ verified on the app.

*Both brackets are evaluated first, then multiplied.*

### Example 5.4 — Squares

Calculate $12^2$. Press **Reset** first.

#32 1 → #33 2 → #24 $x^2$ → #6

Display shows 144 — expected 144. ✔ verified on the app.

*$x^2$ squares the value — quicker and safer than typing the number twice.*

### Example 5.5 — Square roots

Calculate **√144**. Press **Reset** first.

#12 ctrl → #24 √ → #32 1 → #25 4 → #25 4 → #6

Display shows 12 — expected 12. ✔ verified on the app.

*The square-root key returns the principal (positive) root.*

### Example 5.6 — Powers

Calculate **2^10**. Press **Reset** first.

`#33` 2 → `#23` ^ → `#32` 1 → `#39` 0 → `#6`

Display shows `1024` — expected 1024. ✔ verified on the app.

*The power key raises the first number to the second.*

### Example 5.7 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

`#12` ctrl → `#31` log → `#32` 1 → `#39` 0 → `#39` 0 → `#39` 0 → `#38` ) → `#6`

Display shows `3` — expected 3. ✔ verified on the app.

*log is base-10: it counts the zeros of a power of ten.*

### Example 5.8 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

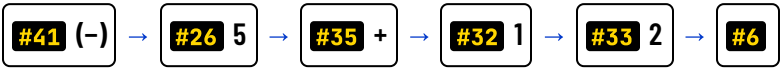
`#12` ctrl → `#30` ln → `#32` 1 → `#38` ) → `#6`

Display shows `0` — expected 0. ✔ verified on the app.

*ln is the natural (base-e) logarithm.*

### Example 5.9 — Negative numbers

**Calculate −5 + 12. Press Reset first.**

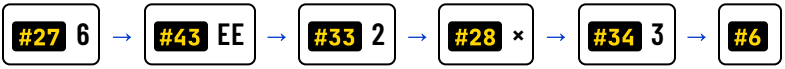



**Display shows** 7 **— expected 7. ✔ verified on the app.**

*Use the sign key for a negative number — not the subtraction key.*

### Example 5.10 — Scientific notation

**Calculate 6×10^2 × 3. Press Reset first.**




**Display shows** 1800 **— expected 1800. ✔ verified on the app.**

*The exponent key enters ×10ⁿ without typing × 10 ^.*

### Example 5.11 — Area of a circle πr²

**Calculate π × 2.5². Press Reset first.**




**Display shows** 19.63495408 **— expected 19.63495408. ✔ verified on the app.**

*π is stored to full precision — never type 3.14.*

### Example 5.12 — Continue from the answer

**Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.**

#32 1 → #39 0 → #39 0 → #28 × → #25 4 → #6 → #29 ÷ → #33 2 → #6

**Display shows** 200 **— expected 200. ✔ verified on the app.**

*Pressing an operator straight after a result continues from that answer (Ans).*

### Example 5.13 — Average of three numbers

**Calculate (4 + 8 + 15) ÷ 3. Press Reset first.**

#37 ( → #25 4 → #35 + → #19 8 → #35 + → #32 1 → #26 5 → #38 ) → #29 ÷ → #34 3 → #6

**Display shows** 9 **— expected 9. ✔ verified on the app.**

*Add up the values, then divide by how many there are.*

### Example 5.14 — Mixed operations

**Calculate 50 − 4 × 6 + 7. Press Reset first.**

#26 5 → #39 0 → #36 − → #25 4 → #28 × → #27 6 → #35 + → #18 7 → #6

**Display shows** 33 **— expected 33. ✔ verified on the app.**

*Only the × part is done first; the − and + are then worked left to right.*

### Example 5.15 — Dividing twice

**Calculate 360 ÷ 4 ÷ 9. Press Reset first.**

#34 3 → #27 6 → #39 0 → #29 ÷ → #25 4 → #29 ÷ → #20 9 → #6

**Display shows** 10 **— expected 10. ✔ verified on the app.**

*Division is worked left to right: the first quotient is divided again.*

### Example 5.16 — Brackets inside a longer formula

**Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.**

#37 ( → #37 ( → #34 3 → #35 + → #26 5 → #38 ) → #28 × → #25 4 → #36 − → #33 2 → #38 ) → #29 ÷ → #32 1 → #39 0 → #6

**Display shows** 3 **— expected 3. ✔ verified on the app.**

*The innermost brackets are worked out first, then the outer brackets, then the division.*

# 6 · Tricks & hacks

### Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

### Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

### Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

### Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

### Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

### Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

### Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

### Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

### Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

### Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

### Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

### Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

### Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

### Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

### Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

### Hack 16: Hidden functions on this model

17 extra functions sit on shifted layers. Press the layer key once, then the key: for example #12 then #7 → off; #12 then #11 → context; #12 then #14 → sto→; #12 then #15 → clear.

### Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #75.

### Hack 18: Type faster on a computer

26 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

### Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

### Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

### Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

# 7 · Keyboard shortcuts

**With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.**

| Keyboard | # | Key | Does |
|---|---|---|---|
| `ArrowDown` | #10 | tab | Cursor ▼ — down / next history line / next item |
| `Backspace` `Delete` | #15 | del | DEL — delete the character at the cursor |
| `s` | #17 | trig | Sine (uses the current angle unit) |
| `7` | #18 | 7 | Digit 7 |
| `8` | #19 | 8 | Digit 8 |
| `9` | #20 | 9 | Digit 9 |
| `^` | #23 | ^ | Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512) |
| `q` | #24 | $x^2$ | Square $x^2$ |
| `4` | #25 | 4 | Digit 4 |
| `5` | #26 | 5 | Digit 5 |
| `6` | #27 | 6 | Digit 6 |
| `*` `x` | #28 | × | Multiplication × |
| `/` | #29 | ÷ | Division ÷ |
| `1` | #32 | 1 | Digit 1 |
| `2` | #33 | 2 | Digit 2 |
| `3` | #34 | 3 | Digit 3 |
| `+` | #35 | + | Addition + |
| `-` | #36 | − | Subtraction − |
| `(` | #37 | ( | Opening bracket ( |
| `)` | #38 | ) | Closing bracket ). Closing brackets at the end of a line may be omitted |

| | | | |
|---|---|---|---|
| 0 | #39 | 0 | Digit 0 |
| . | #40 | . | Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored |
| Enter = | #42 | enter | Execute / = — evaluate the line and show the result (EXE / ENTER / =) |
| e | #43 | EE | ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23 |
| p | #52 | π► | π (3.14159265…) as an exact constant |
| , | #61 | , | Comma / argument separator (between function arguments, list items or coordinates) |

## App shortcuts

| | |
|---|---|
| “# On / # Off” | show / hide key numbers |
| “Classic / Hi-Contrast” | switch key colours |
| “Copy” | copy display text |
| “Fit / 1:1” | scale to screen / design size |
| “Reset” | restart this calculator |
| Jump list | switch to another of the 45 calculators |

# 8 · Common mistakes

**Mistake 1: (–) vs –**

Use the negative key for negative numbers.

**Mistake 2: Radian mode**

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

**Mistake 3: Implicit multiplication**

2/3x may be read as 2/(3x) depending on the model; use brackets.

**Mistake 4: Using – instead of (–)**

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

**Mistake 5: Wrong angle unit**

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

**Mistake 6: $-3^2$ vs $(-3)^2$**

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

### Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

### Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

### Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

### Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

### Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

### Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

# 9 · Practice drills

## Set A — Calculation drills (89)

**Work each one on calculator #17, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.**

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. $25^2$ = _____

21. $7^2$ = _____

22. $1.5^2$ = _____

23. $19^2$ = _____

24. $33^2$ = _____

25. $0.5^2$ = _____

26. $125^2$ = _____

27. $64^2$ = _____

28. √625 = _____

29. √1024 = _____

30. √9801 = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

**37.** $2^{0.5}$ = _____

**38.** $\log 100$ = _____

**39.** $\log 10000$ = _____

**40.** $\ln 10$ = _____

**41.** $\ln 2$ = _____

**42.** $-20 + 7$ = _____

**43.** $-3 + 3$ = _____

**44.** $-100 + 250$ = _____

**45.** $-8 + 1$ = _____

**46.** $2\times10^{3} \times 4$ = _____

**47.** $1.5\times10^{4} \times 2$ = _____

**48.** $3\times10^{5} \times 3$ = _____

**49.** $4.5\times10^{2} \times 2$ = _____

**50.** $\pi \times 1.2^{2}$ = _____

**51.** $15 \times 6$ = , then Ans $\div 9$ = _____

**52.** $7.5 \times 8$ = , then Ans $\div 3$ = _____

**53.** $250 \times 12$ = , then Ans $\div 100$ = _____

**54.** $(12 + 15 + 18) \div 3$ = _____

**55.** $(70 + 85 + 91) \div 3$ = _____

**56.** $(2.5 + 3.5 + 6) \div 3$ = _____

57. (100 + 200 + 600) ÷ 3 = _____

58. 100 − 3 × 12 + 9 = _____

59. 18 − 2 × 5 + 40 = _____

60. 7.5 − 1.5 × 3 + 2 = _____

61. 64 − 8 × 4 + 10 = _____

62. 300 − 25 × 4 + 55 = _____

63. 9 − 3 × 3 + 1 = _____

64. 45 − 5 × 5 + 15 = _____

65. 1000 ÷ 8 ÷ 5 = _____

66. 96 ÷ 2 ÷ 3 = _____

67. 7200 ÷ 60 ÷ 60 = _____

68. 500 ÷ 4 ÷ 25 = _____

69. 81 ÷ 3 ÷ 3 = _____

70. ((10 + 2) × 3 − 6) ÷ 6 = _____

71. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

72. ((20 + 30) × 2 − 25) ÷ 3 = _____

73. ((7 + 8) × 9 − 15) ÷ 12 = _____

74. 12 + 3 × 4 = _____

75. (12 + 8) ÷ 4 = _____

76. (3 + 4) × (9 − 2) = _____

77. $12^2$ = _____

78. $\sqrt{144}$ = _____

79. 2^10 = _____

80. log 1000 = _____

81. ln 1 = _____

82. −5 + 12 = _____

83. 6×10^2 × 3 = _____

84. π × $2.5^2$ = _____

85. 100 × 4 = , then Ans ÷ 2 = _____

86. (4 + 8 + 15) ÷ 3 = _____

87. 50 − 4 × 6 + 7 = _____

88. 360 ÷ 4 ÷ 9 = _____

89. ((3 + 5) × 4 − 2) ÷ 10 = _____

## Set B — Key-finding drills (78)

**Without looking at the table, find each key on the picture. Answers in Appendix A.**

1. Which key is “esc” (QUIT)? _____
2. Which key is “▲” (Cursor ▲)? _____
3. Which key is “▼” (Cursor ▼)? _____
4. Which key is “◀” (Cursor ◀)? _____
5. Which key is “▶” (Cursor ▶)? _____

6. Which key is “exe” (Execute / =)? _____

7. Which key is “⌂ on” (APPS)? _____

8. Which key is “🗒” (QUIT)? _____

9. Which key is “doc” (APPS)? _____

10. Which key is “tab” (Cursor ▼)? _____

11. Which key is “menu” (MATH menu)? _____

12. Which key is “ctrl” (SHIFT / 2nd)? _____

13. Which key is “var” (VARS)? _____

14. Which key is “del” (DEL)? _____

15. Which key is “=” (Equals)? _____

16. Which key is “trig” (Sine)? _____

17. Which key is “▭▭” (Fraction template ▭/▭)? _____

18. Which key is “📖” (CATALOG)? _____

19. Which key is “^” (Power $x^y$ / ^)? _____

20. Which key is “$x^2$” (Square $x^2$)? _____

21. Which key is “×” (Multiplication ×)? _____

22. Which key is “÷” (Division ÷)? _____

23. Which key is “ex” ($e^x$)? _____

24. Which key is “10x” ($10^x$)? _____

25. Which key is “+” (Addition +)? _____

**26. Which key is "−" (Subtraction −)? _____**

**27. Which key is "(" (Opening bracket)? _____**

**28. Which key is ")" (Closing bracket ). Closing brackets at the end of a line may be omitted)? _____**

**29. Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____**

**30. Which key is "(−)" (Negative sign)? _____**

**31. Which key is "enter" (Execute / =)? _____**

**32. Which key is "EE" (×10ˣ / EXP / EE)? _____**

**33. Which key is "A" (Variable A)? _____**

**34. Which key is "B" (Variable B)? _____**

**35. Which key is "C" (Variable C)? _____**

**36. Which key is "D" (Variable D)? _____**

**37. Which key is "E" (Variable E)? _____**

**38. Which key is "F" (Variable F)? _____**

**39. Which key is "G" (Variable G)? _____**

**40. Which key is "π▶" (π)? _____**

**41. Which key is "H" (Variable H)? _____**

**42. Which key is "I" (Variable I)? _____**

**43. Which key is "J" (Variable J)? _____**

**44. Which key is "K" (Variable K)? _____**

**45. Which key is “L” (Variable L)? _____**

**46. Which key is “M” (Variable M)? _____**

**47. Which key is “N” (Variable N)? _____**

**48. Which key is “,” (Comma / argument separator)? _____**

**49. Which key is “O” (Variable O)? _____**

**50. Which key is “P” (Variable P)? _____**

**51. Which key is “Q” (Variable Q)? _____**

**52. Which key is “R” (Variable R)? _____**

**53. Which key is “S” (Variable S)? _____**

**54. Which key is “T” (Variable T)? _____**

**55. Which key is “U” (Variable U)? _____**

**56. Which key is “↵” (Execute / =)? _____**

**57. Which key is “V” (Variable V)? _____**

**58. Which key is “W” (Variable W)? _____**

**59. Which key is “X” (Variable X)? _____**

**60. Which key is “Y” (Variable Y)? _____**

**61. Which key is “Z” (Variable Z)? _____**

**62. How do you reach “off”? _____**

**63. How do you reach “context”? _____**

**64. How do you reach “sto →”? _____**

**65. How do you reach “clear”? _____**

**66. How do you reach “≠≥►”? _____**

**67. How do you reach “$\sin^{-1}$”? _____**

**68. How do you reach “∞β°”? _____**

**69. How do you reach “Σ”? _____**

**70. How do you reach “n√”? _____**

**71. How do you reach “√”? _____**

**72. How do you reach “/”? _____**

**73. How do you reach “ln”? _____**

**74. How do you reach “log”? _____**

**75. How do you reach “[ ]”? _____**

**76. How do you reach “{ }”? _____**

**77. How do you reach “ans”? _____**

**78. How do you reach “≈”? _____**

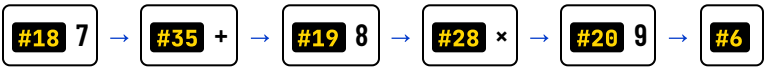
# Appendix A — Drill answers

## Set A

1. **55** (display `55`)




2. **79** (display `79`)



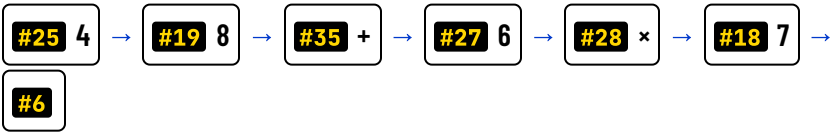

3. **282** (display `282`)




4. **213** (display `213`)




5. **90** (display `90`)




6. **1100** (display `1100`)



**7. 82 (display 82)**




**8. 128 (display 128)**



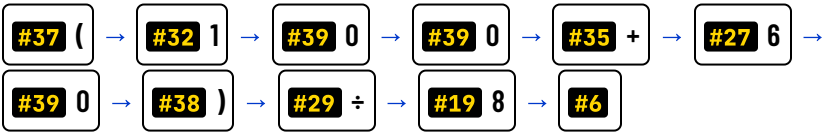

**9. 10 (display 10)**




**10. 8 (display 8)**




**11. 20 (display 20)**




**12. 10 (display 10)**



### 13. 50 (display 50)

#37 ( → #33 2 → #26 5 → #39 0 → #35 + → #34 3 → #26 5 → #39 0 → #38 ) → #29 ÷ → #32 1 → #33 2 → #6

### 14. 77 (display 77)

#37 ( → #26 5 → #35 + → #27 6 → #38 ) → #28 × → #37 ( → #32 1 → #39 0 → #36 − → #34 3 → #38 ) → #6

### 15. 45 (display 45)

#37 ( → #32 1 → #33 2 → #35 + → #34 3 → #38 ) → #28 × → #37 ( → #19 8 → #36 − → #26 5 → #38 ) → #6

### 16. 12 (display 12)

#37 ( → #32 1 → #40 . → #26 5 → #35 + → #33 2 → #40 . → #26 5 → #38 ) → #28 × → #37 ( → #18 7 → #36 − → #25 4 → #38 ) → #6

### 17. 500 (display 500)

#37 ( → #33 2 → #39 0 → #35 + → #34 3 → #39 0 → #38 ) → #28 × → #37 ( → #32 1 → #26 5 → #36 − → #26 5 → #38 ) → #6

**18. 990** (display `990`)

`#37` ( → `#20` 9 → `#35` + → `#32` 1 → `#38` ) → `#28` × →
`#37` ( → `#32` 1 → `#39` 0 → `#39` 0 → `#36` − → `#32` 1 →
`#38` ) → `#6`

**19. 192** (display `192`)

`#37` ( → `#32` 1 → `#32` 1 → `#35` + → `#32` 1 → `#34` 3 →
`#38` ) → `#28` × → `#37` ( → `#32` 1 → `#18` 7 → `#36` − →
`#20` 9 → `#38` ) → `#6`

**20. 625** (display `625`)

`#33` 2 → `#26` 5 → `#24` $x^2$ → `#6`

**21. 49** (display `49`)

`#18` 7 → `#24` $x^2$ → `#6`

**22. 2.25** (display `2.25`)

`#32` 1 → `#40` . → `#26` 5 → `#24` $x^2$ → `#6`

**23. 361** (display `361`)

`#32` 1 → `#20` 9 → `#24` $x^2$ → `#6`

**24. 1089** (display `1089`)

`#34` 3 → `#34` 3 → `#24` $x^2$ → `#6`

**25. 0.25** (display `0.25`)

`#39` 0 → `#40` . → `#26` 5 → `#24` $x^2$ → `#6`

**26. 15625** (display `15625`)

`#32` 1 → `#33` 2 → `#26` 5 → `#24` $x^2$ → `#6`

**27. 4096** (display `4096`)

`#27` 6 → `#25` 4 → `#24` $x^2$ → `#6`

**28. 25** (display `25`)

`#12` ctrl → `#24` √ → `#27` 6 → `#33` 2 → `#26` 5 → `#6`

**29. 32** (display `32`)

`#12` ctrl → `#24` √ → `#32` 1 → `#39` 0 → `#33` 2 → `#25` 4 → `#6`

**30. 99** (display `99`)

`#12` ctrl → `#24` √ → `#20` 9 → `#19` 8 → `#39` 0 → `#32` 1 → `#6`

**31. 243** (display `243`)

`#34` 3 → `#23` ^ → `#26` 5 → `#6`

**32. 1.157625** (display `1.157625`)

`#32` 1 → `#40` . → `#39` 0 → `#26` 5 → `#23` ^ → `#34` 3 → `#6`

**33. 625** (display `625`)

`#26` 5 → `#23` ^ → `#25` 4 → `#6`

**34. 343** (display `343`)

`#18` 7 → `#23` ^ → `#34` 3 → `#6`

**35. 2.25** (display `2.25`)

#32 1 → #40 . → #26 5 → #23 ^ → #33 2 → #6

**36. 1000000.000** (display `1000000`)

#32 1 → #39 0 → #23 ^ → #27 6 → #6

**37. 1.41421356** (display `1.414213562`)

#33 2 → #23 ^ → #39 0 → #40 . → #26 5 → #6

**38. 2** (display `2`)

#12 ctrl → #31 log → #32 1 → #39 0 → #39 0 → #38 ) → #6

**39. 4** (display `4`)

#12 ctrl → #31 log → #32 1 → #39 0 → #39 0 → #39 0 → #39 0 → #38 ) → #6

**40. 2.30258509** (display `2.302585093`)

#12 ctrl → #30 ln → #32 1 → #39 0 → #38 ) → #6

**41. 0.69314718** (display `0.6931471806`)

#12 ctrl → #30 ln → #33 2 → #38 ) → #6

**42. -13** (display `-13`)

#41 (–) → #33 2 → #39 0 → #35 + → #18 7 → #6

**43. 0** (display `0`)

#41 (–) → #34 3 → #35 + → #34 3 → #6

**44. 150** (display `150`)

#41 (−) → #32 1 → #39 0 → #39 0 → #35 + → #33 2 → #26 5 → #39 0 → #6

**45. -7** (display `-7`)

#41 (−) → #19 8 → #35 + → #32 1 → #6

**46. 8000** (display `8000`)

#33 2 → #43 EE → #34 3 → #28 × → #25 4 → #6

**47. 30000** (display `30000`)

#32 1 → #40 . → #26 5 → #43 EE → #25 4 → #28 × → #33 2 → #6

**48. 900000** (display `900000`)

#34 3 → #43 EE → #26 5 → #28 × → #34 3 → #6

**49. 900** (display `900`)

#25 4 → #40 . → #26 5 → #43 EE → #33 2 → #28 × → #33 2 → #6

**50. 4.52389342** (display `4.523893421`)

#52 π▶ → #28 × → #32 1 → #40 . → #33 2 → #24 $x^2$ → #6

**51. 10** (display `10`)

#32 1 → #26 5 → #28 × → #27 6 → #6 → #29 ÷ → #20 9 → #6

**52. 20** (display `20`)

`#18` 7 → `#40` . → `#26` 5 → `#28` × → `#19` 8 → `#6` → `#29` ÷ → `#34` 3 → `#6`

**53. 30** (display `30`)

`#33` 2 → `#26` 5 → `#39` 0 → `#28` × → `#32` 1 → `#33` 2 → `#6` → `#29` ÷ → `#32` 1 → `#39` 0 → `#39` 0 → `#6`

**54. 15** (display `15`)

`#37` ( → `#32` 1 → `#33` 2 → `#35` + → `#32` 1 → `#26` 5 → `#35` + → `#32` 1 → `#19` 8 → `#38` ) → `#29` ÷ → `#34` 3 → `#6`

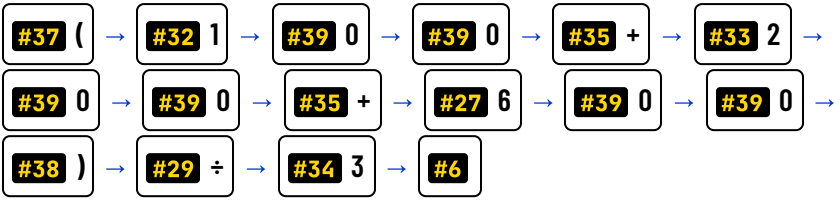
**55. 82** (display `82`)

`#37` ( → `#18` 7 → `#39` 0 → `#35` + → `#19` 8 → `#26` 5 → `#35` + → `#20` 9 → `#32` 1 → `#38` ) → `#29` ÷ → `#34` 3 → `#6`

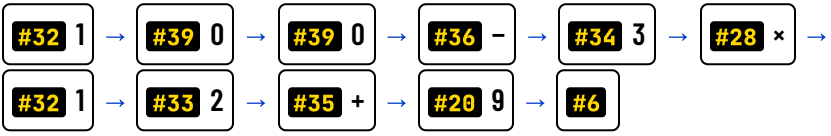
**56. 4** (display `4`)

`#37` ( → `#33` 2 → `#40` . → `#26` 5 → `#35` + → `#34` 3 → `#40` . → `#26` 5 → `#35` + → `#27` 6 → `#38` ) → `#29` ÷ → `#34` 3 → `#6`

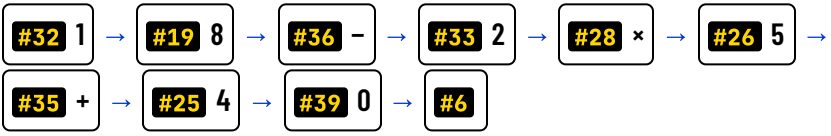
**57. 300 (display 300)**




**58. 73 (display 73)**




**59. 48 (display 48)**



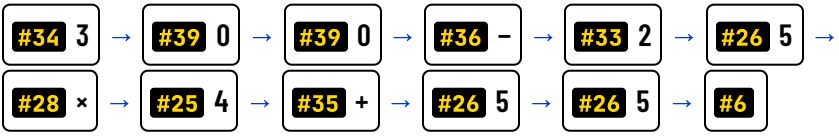

**60. 5 (display 5)**




**61. 42 (display 42)**




**62. 255 (display 255)**



**63. 1 (display 1)**




**64. 35 (display 35)**




**65. 25 (display 25)**



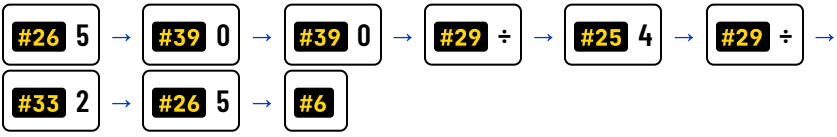

**66. 16 (display 16)**




**67. 2 (display 2)**




**68. 5 (display 5)**




**69. 9 (display 9)**



### 70. 5 (display 5)

#37 ( → #37 ( → #32 1 → #39 0 → #35 + → #33 2 → #38 ) → #28 × → #34 3 → #36 − → #27 6 → #38 ) → #29 ÷ → #27 6 → #6

### 71. 4 (display 4)

#37 ( → #37 ( → #32 1 → #40 . → #26 5 → #35 + → #33 2 → #40 . → #26 5 → #38 ) → #28 × → #27 6 → #36 − → #25 4 → #38 ) → #29 ÷ → #26 5 → #6

### 72. 25 (display 25)

#37 ( → #37 ( → #33 2 → #39 0 → #35 + → #34 3 → #39 0 → #38 ) → #28 × → #33 2 → #36 − → #33 2 → #26 5 → #38 ) → #29 ÷ → #34 3 → #6

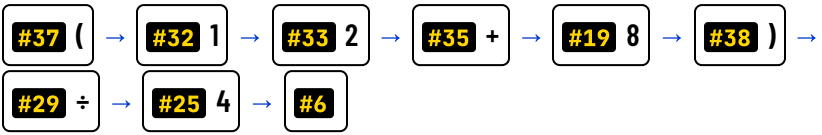
### 73. 10 (display 10)

#37 ( → #37 ( → #18 7 → #35 + → #19 8 → #38 ) → #28 × → #20 9 → #36 − → #32 1 → #26 5 → #38 ) → #29 ÷ → #32 1 → #33 2 → #6

### 74. 24 (display 24)

#32 1 → #33 2 → #35 + → #34 3 → #28 × → #25 4 → #6

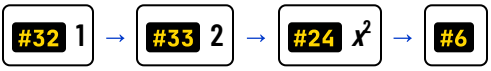
**75. 5** (display `5`)




**76. 49** (display `49`)




**77. 144** (display `144`)




**78. 12** (display `12`)




**79. 1024** (display `1024`)



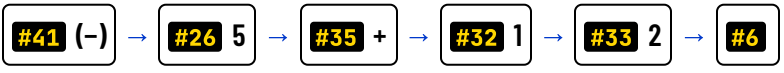

**80. 3** (display `3`)



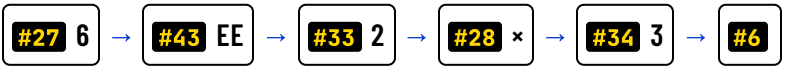

**81. 0** (display `0`)




**82. 7** (display `7`)




**83. 1800** (display `1800`)



**84. 19.63495408** (display `19.63495408`)

#52 π▶ → #28 × → #33 2 → #40 . → #26 5 → #24 $x^2$ → #6

**85. 200** (display `200`)

#32 1 → #39 0 → #39 0 → #28 × → #25 4 → #6 → #29 ÷ → #33 2 → #6

**86. 9** (display `9`)

#37 ( → #25 4 → #35 + → #19 8 → #35 + → #32 1 → #26 5 → #38 ) → #29 ÷ → #34 3 → #6

**87. 33** (display `33`)

#26 5 → #39 0 → #36 − → #25 4 → #28 × → #27 6 → #35 + → #18 7 → #6

**88. 10** (display `10`)

#34 3 → #27 6 → #39 0 → #29 ÷ → #25 4 → #29 ÷ → #20 9 → #6

**89. 3** (display `3`)

#37 ( → #37 ( → #34 3 → #35 + → #26 5 → #38 ) → #28 × → #25 4 → #36 − → #33 2 → #38 ) → #29 ÷ → #32 1 → #39 0 → #6

## Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #14
14. #15
15. #16
16. #17
17. #21
18. #22
19. #23
20. #24
21. #28
22. #29
23. #30
24. #31
25. #35
26. #36
27. #37
28. #38
29. #40
30. #41
31. #42
32. #43
33. #44
34. #45
35. #46
36. #47
37. #48
38. #49
39. #50
40. #52
41. #53
42. #54
43. #55
44. #56
45. #57
46. #58
47. #59
48. #61
49. #62
50. #63
51. #64
52. #65
53. #66
54. #67
55. #68
56. #69
57. #70
58. #71
59. #72
60. #73
61. #74
62. #12 ctrl then #7 ⌂ on
63. #12 ctrl then #11 menu
64. #12 ctrl then #14 var
65. #12 ctrl then #15 del
66. #12 ctrl then #16 =
67. #12 ctrl then #17 trig
68. #12 ctrl then #21 ▭▭
69. #12 ctrl then #22 📖
70. #12 ctrl then #23 ^
71. #12 ctrl then #24 $x^2$
72. #12 ctrl then #29 ÷
73. #12 ctrl then #30 ex
74. #12 ctrl then #31 10x
75. #12 ctrl then #37 (
76. #12 ctrl then #38 )
77. #12 ctrl then #41 (-)
78. #12 ctrl then #42 enter

# Appendix B — Function finder (A–Z)

| Function | Label | Keys |
|---|---|---|
| ×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23 | EE | #43 |
| ▶Decimal — show the result as a decimal | ≈ | #12 → #42 |
| 10^x — antilog | 10x | #31 |
| Addition + | + | #35 |
| Ans — recall the last answer into the current expression | ans | #12 → #41 |
| APPS — applications | ⌂ on | #7 |
| APPS — applications | doc | #9 |
| CATALOG — list of all commands | 📖 | #22 |
| CLEAR — clear the entry line / screen | clear | #12 → #15 |
| Closing bracket ). Closing brackets at the end of a line may be omitted | ) | #38 |
| Comma / argument separator (between function arguments, list items or coordinates) | , | #61 |
| Common logarithm $\log_{10}$ | log | #12 → #31 |
| Cursor ▲ — up / previous history line / previous item | ▲ | #2 |
| Cursor ▶ — move right in the line | ▶ | #5 |
| Cursor ▼ — down / next history line / next item | ▼ | #3 |
| Cursor ▼ — down / next history line / next item | tab | #10 |
| Cursor ◀ — move left in the line | ◀ | #4 |
| Decimal point. Pressing it first gives "0."; a second point in the same number is ignored | . | #40 |
| Degrees-minutes-seconds entry/display °′ ″ | ∞β° | #12 → #21 |
| DEL — delete the character at the cursor | del | #15 |

| | | |
|---|---|---|
| Digit 0 | 0 | #39 |
| Digit 1 | 1 | #32 |
| Digit 2 | 2 | #33 |
| Digit 3 | 3 | #34 |
| Digit 4 | 4 | #25 |
| Digit 5 | 5 | #26 |
| Digit 6 | 6 | #27 |
| Digit 7 | 7 | #18 |
| Digit 8 | 8 | #19 |
| Digit 9 | 9 | #20 |
| Division ÷ | ÷ | #29 |
| Equals — complete the calculation | = | #16 |
| $e^x$ — exponential | ex | #30 |
| Execute / = — evaluate the line and show the result (EXE / ENTER / =) | exe | #6 |
| Execute / = — evaluate the line and show the result (EXE / ENTER / =) | enter | #42 |
| Execute / = — evaluate the line and show the result (EXE / ENTER / =) | ↵ | #69 |
| Fraction template ▭/▭ (numerator over denominator) | ▭▭ | #21 |
| Fraction template ▭/▭ (numerator over denominator) | / | #12 → #29 |
| Inverse sine $\sin^{-1}$ | $\sin^{-1}$ | #12 → #17 |
| MATH menu — extra math functions | menu | #11 |
| Multiplication × | × | #28 |
| n-th root $\sqrt[x]{y}$ | $\sqrt[n]{}$ | #12 → #23 |
| Natural logarithm ln | ln | #12 → #30 |
| Negative sign (−) — makes the next number negative. Different from the subtraction key | (−) | #41 |
| OFF — power off (state is lost; press ON to restart) | off | #12 → #7 |

| | | |
|---|---|---|
| Opening bracket ( | ( | #37 |
| OPTN — options for the current mode | context | #12 → #11 |
| Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512) | ^ | #23 |
| Printed label only (no action in this app) | shift | #13 |
| Printed label only (no action in this app) | ?!▸ | #51 |
| Printed label only (no action in this app) | ⚑ | #60 |
| Printed label only (no action in this app) | ␣ | #75 |
| QUIT — leave the current screen / app | esc | #1 |
| QUIT — leave the current screen / app | 🗒 | #8 |
| SHIFT / 2nd — the next key uses its upper (shift-coloured) function | ctrl | #12 |
| Sine (uses the current angle unit) | trig | #17 |
| Square root √ | √ | #12 → #24 |
| Square $x^2$ | $x^2$ | #24 |
| STO▶ ( → ) — store into a variable | sto → | #12 → #14 |
| Subtraction − | − | #36 |
| Summation Σ over an integer range | Σ | #12 → #22 |
| Types "[" | [ ] | #12 → #37 |
| Types "{" | { } | #12 → #38 |
| Types "≠" | ≠≥▸ | #12 → #16 |
| Variable A (store with STO/ →, recall by name) | A | #44 |
| Variable B (store with STO/ →, recall by name) | B | #45 |
| Variable C (store with STO/ →, recall by name) | C | #46 |
| Variable D (store with STO/ →, recall by name) | D | #47 |

| | | |
|---|---|---|
| Variable E (store with STO/ →, recall by name) | E | #48 |
| Variable F (store with STO/ →, recall by name) | F | #49 |
| Variable G (store with STO/ →, recall by name) | G | #50 |
| Variable H (store with STO/ →, recall by name) | H | #53 |
| Variable I (store with STO/ →, recall by name) | I | #54 |
| Variable J (store with STO/ →, recall by name) | J | #55 |
| Variable K (store with STO/ →, recall by name) | K | #56 |
| Variable L (store with STO/ →, recall by name) | L | #57 |
| Variable M (store with STO/ →, recall by name) | M | #58 |
| Variable N (store with STO/ →, recall by name) | N | #59 |
| Variable O (store with STO/ →, recall by name) | O | #62 |
| Variable P (store with STO/ →, recall by name) | P | #63 |
| Variable Q (store with STO/ →, recall by name) | Q | #64 |
| Variable R (store with STO/ →, recall by name) | R | #65 |
| Variable S (store with STO/ →, recall by name) | S | #66 |
| Variable T (store with STO/ →, recall by name) | T | #67 |
| Variable U (store with STO/ →, recall by name) | U | #68 |
| Variable V (store with STO/ →, recall by name) | V | #70 |
| Variable W (store with STO/ →, recall by name) | W | #71 |
| Variable X (store with STO/ →, recall by name) | X | #72 |
| Variable Y (store with STO/ →, recall by name) | Y | #73 |
| Variable Z (store with STO/ →, recall by name) | Z | #74 |
| VARS — variables menu | var | #14 |
| π (3.14159265…) as an exact constant | π▶ | #52 |

# Appendix C — Glossary

**Ans**
The result of the last calculation, reusable in the next.

**AOS**
Algebraic Operating System — operator precedence on a one-line display.

**RPN**
Reverse Polish Notation — operands first, then the operation; no = key.

**Stack**
The RPN registers X, Y, Z, T where numbers wait for operations.

**TVM**
Time Value of Money — n, i, PV, PMT, FV.

**Natural display**
Shows fractions, roots and powers as printed in textbooks.

**Shift / 2nd**
Prefix key that selects the function printed above a key.

**Alpha**
Prefix key that types the letter printed beside a key.

**Radian**
Angle unit; 2π rad = 360°.

**Grad**
Angle unit; 400 grad = 360°.

**Engineering notation**
Scientific notation with exponents that are multiples of 3.

**Mantissa**
The digits of a number in scientific notation.

**Memory (M)**

An independent register for running totals.

**Mark-up**

Increase over cost; margin is profit as a percent of selling price.

**Grand total (GT)**

Sum of all totals since the last clear.

**nCr / nPr**

Combinations / permutations of n items taken r at a time.

**Factorial**

n! = 1×2×…×n.

**Reciprocal**

1/x.

**Standard deviation**

Spread of data around the mean (σ population, s sample).

**Key number**

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

# Appendix D — Model notes & limits

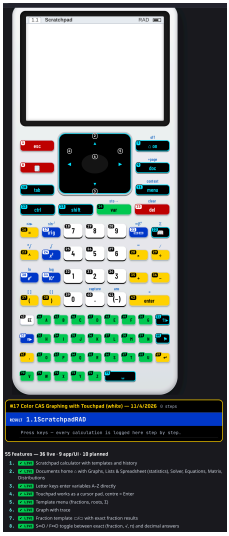
Touchpad CAS-style colour model: Scratchpad calculator with templates and history, documents home (⌂) with Graphs, Lists & Spreadsheet (statistics), Solver, Equations, Matrix, Distributions. Letter keys enter variables A–Z directly. The trig key types sin (ctrl gives $\sin^{-1}$; cos/tan via the menu). Symbolic CAS algebra, documents/pages, Python and the touchpad gestures are simplified (the touchpad works as a cursor pad, its centre as Enter).

## Planned (not yet working)

- Symbolic CAS algebra
- Documents / pages
- Python
- Touchpad gestures
- Geometry
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

# Appendix E — Quick-reference card



| | | |
|---|---|---|
| #1 esc | #2 ▲ | #3 ▼ |
| #4 ◀ | #5 ▶ | #6 exe |
| #7 ⌂ on ⇧ off | #8 🗒 | #9 doc ⇧ +page |
| #10 tab | #11 menu ⇧ context | #12 ctrl |
| #13 shift | #14 var ⇧ sto→ | #15 del ⇧ clear |
| #16 = ⇧ ≠≥▶ | #17 trig ⇧ $\sin^{-1}$ | #18 7 |
| #19 8 | #20 9 | #21 ▭▭ ⇧ ∞β° |
| #22 📖 ⇧ Σ | #23 ^ ⇧ $\sqrt[n]{}$ | #24 $x^2$ ⇧ √ |
| #25 4 | #26 5 | #27 6 |
| #28 × ⇧ "" | #29 ÷ ⇧ / | #30 ex ⇧ ln |
| #31 10x ⇧ log | #32 1 | #33 2 |
| #34 3 | #35 + | #36 – |
| #37 ( ⇧ [ ] | #38 ) ⇧ { } | #39 0 |
| #40 . ⇧ capture | #41 (–) ⇧ ans | #42 enter ⇧ ≈ |
| #43 EE | #44 A | #45 B |
| #46 C | #47 D | #48 E |
| #49 F | #50 G | #51 ?!▶ |

| | | |
|---|---|---|
| #52 π► | #53 H | #54 I |
| #55 J | #56 K | #57 L |
| #58 M | #59 N | #60 ⚑ |
| #61 , | #62 O | #63 P |
| #64 Q | #65 R | #66 S |
| #67 T | #68 U | #69 ↵ |
| #70 V | #71 W | #72 X |
| #73 Y | #74 Z | #75 ␣ |

U-45 CALCULATORS · Calculator #17 · cut out and keep